

The Group Spencer Conjecture for Finite Groups

Literature identification for arXiv:2212.00066

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Status. literature_already_answered.

Source. A. S. Bandeira, D. Kunisky, D. G. Mixon, and X. Zeng, *On the concentration of Gaussian Cayley matrices*, arXiv:2212.00066; Applied and Computational Harmonic Analysis 73 (2024), 101694 [1].

For a finite group G , let ρ be its left regular representation. The source proves in Theorem 12 (PDF page 8) that, when G is abelian or simple, there are signs $\varepsilon_g \in \{-1, 1\}$ such that

$$\left\| \sum_{g \in G} \varepsilon_g \rho(g) \right\| \lesssim \sqrt{|G|}.$$

Its Discussion (PDF page 9) then states that it would be interesting to prove this theorem for more general groups and records the obstruction to iterating the authors' partial-coloring argument.

Supporting answer. A. S. Bandeira and H. Boelskei, *Matrix Discrepancy for Representations of Finite Groups*, arXiv:2606.12181v2 [2].

The supporting authors explicitly identify the general finite-group case as the conjecture proposed by the source and prove the following as their Theorem 2 (PDF page 2): there is a universal constant C such that, for every finite group G ,

$$\min_{\varepsilon \in \{\pm 1\}^G} \left\| \sum_{g \in G} \varepsilon_g \rho(g) \right\| \leq C \sqrt{|G|}.$$

This is exactly the all-finite-groups extension requested in the source. The answer paper also explains that it overcomes the stated iteration obstacle by combining partial coloring with intrinsic-freeness inequalities.

Scope. The identification settles the Group Spencer conjecture for left regular representations (indeed, the answer notes that the theorem holds for any unitary representation of a finite group). It does not settle the general Matrix Spencer conjecture for arbitrary families of contraction matrices, which the supporting paper expressly leaves open.

Search evidence. Exact searches for “Matrix Spencer conjecture”, the source title, and finite-group representations located arXiv:2606.12181. That paper cites the source as [BKM+24] and says that the source left the general finite-group case as a conjecture. A contemporaneous independent paper, arXiv:2606.16005, proves a broader algebraic special case, but is not needed for this exact identification.

References

- [1] A. S. Bandeira, D. Kunisky, D. G. Mixon, and X. Zeng, *On the concentration of Gaussian Cayley matrices*, Applied and Computational Harmonic Analysis 73 (2024), 101694; arXiv:2212.00066.
- [2] A. S. Bandeira and H. Boelcskei, *Matrix Discrepancy for Representations of Finite Groups*, arXiv:2606.12181v2, 2026.